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```
print(lowercase_paragraph)
```

```
natural language processing (nlp) is a field of artificial intelligence that foc
```

✓ Task

Install NLTK and spaCy libraries, import them, define a sample paragraph, count the words, convert the text to lowercase, and then summarize these preprocessing steps, including the word count and the lowercase version of the text.

✓ Install NLP Libraries

Subtask:

Install the NLTK and spaCy libraries in the Colab environment.

Reasoning: Install the NLTK and spaCy libraries as instructed in the subtask.

```
!pip install nltk
!pip install spacy
print("NLTK and spaCy libraries installed.")
```

```
Requirement already satisfied: nltk in /usr/local/lib/python3.12/dist-packages (
Requirement already satisfied: click in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: joblib in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: regex<=2021.8.3 in /usr/local/lib/python3.12/dist
Requirement already satisfied: tqdm in /usr/local/lib/python3.12/dist-packages (
Requirement already satisfied: spacy in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: spacy-legacy<3.1.0,>=3.0.11 in /usr/local/lib/pyt
Requirement already satisfied: spacy-loggers<2.0.0,>=1.0.0 in /usr/local/lib/pyt
Requirement already satisfied: murmurhash<1.1.0,>=0.28.0 in /usr/local/lib/pytho
Requirement already satisfied: cymem<2.1.0,>=2.0.2 in /usr/local/lib/python3.12/
Requirement already satisfied: preshed<3.1.0,>=3.0.2 in /usr/local/lib/python3.1
Requirement already satisfied: thinc<8.4.0,>=8.3.4 in /usr/local/lib/python3.12/
Requirement already satisfied: wasabi<1.2.0,>=0.9.1 in /usr/local/lib/python3.12
Requirement already satisfied: srsly<3.0.0,>=2.4.3 in /usr/local/lib/python3.12/
Requirement already satisfied: catalogue<2.1.0,>=2.0.6 in /usr/local/lib/python3
Requirement already satisfied: weasel<0.5.0,>=0.4.2 in /usr/local/lib/python3.12
Requirement already satisfied: typer-slim<1.0.0,>=0.3.0 in /usr/local/lib/python
Requirement already satisfied: tqdm<5.0.0,>=4.38.0 in /usr/local/lib/python3.12/
Requirement already satisfied: numpy>=1.19.0 in /usr/local/lib/python3.12/dist-p
```

```
Requirement already satisfied: requests<3.0.0,>=2.13.0 in /usr/local/lib/python3
Requirement already satisfied: pydantic!=1.8,!1.8.1,<3.0.0,>=1.7.4 in /usr/loca
Requirement already satisfied: jinja2 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: setuptools in /usr/local/lib/python3.12/dist-pack
Requirement already satisfied: packaging>=20.0 in /usr/local/lib/python3.12/dist
Requirement already satisfied: annotated-types>=0.6.0 in /usr/local/lib/python3.
Requirement already satisfied: pydantic-core==2.41.4 in /usr/local/lib/python3.1
Requirement already satisfied: typing-extensions>=4.14.1 in /usr/local/lib/pytho
Requirement already satisfied: typing-inspection>=0.4.2 in /usr/local/lib/python
Requirement already satisfied: charset_normalizer<4,>=2 in /usr/local/lib/python
Requirement already satisfied: idna<4,>=2.5 in /usr/local/lib/python3.12/dist-pa
Requirement already satisfied: urllib3<3,>=1.21.1 in /usr/local/lib/python3.12/d
Requirement already satisfied: certifi>=2017.4.17 in /usr/local/lib/python3.12/d
Requirement already satisfied: blis<1.4.0,>=1.3.0 in /usr/local/lib/python3.12/d
Requirement already satisfied: confection<1.0.0,>=0.0.1 in /usr/local/lib/python
Requirement already satisfied: click>=8.0.0 in /usr/local/lib/python3.12/dist-pa
Requirement already satisfied: cloudpathlib<1.0.0,>=0.7.0 in /usr/local/lib/pyth
Requirement already satisfied: smart-open<8.0.0,>=5.2.1 in /usr/local/lib/python
Requirement already satisfied: MarkupSafe>=2.0 in /usr/local/lib/python3.12/dist
Requirement already satisfied: wrapt in /usr/local/lib/python3.12/dist-packages
NLTK and spaCy libraries installed.
```

✓ Import NLP Libraries

Subtask:

Import the installed NLTK and spaCy libraries for use in the notebook.

Reasoning: Import the NLTK and spaCy libraries as instructed in the subtask.

```
import nltk
import spacy
print("NLTK and spaCy libraries imported.")
```

```
NLTK and spaCy libraries imported.
```

✓ Define Sample Text

Subtask:

Load a short paragraph of your choice as a string variable in the notebook. This can be a news snippet or a poem.

Reasoning: The subtask requires defining a sample paragraph as a string variable named `sample_paragraph`. I will create a code cell to declare this variable and assign it

a sample text.

```
sample_paragraph = "Natural Language Processing (NLP) is a field of artificial  
print("Sample paragraph defined.")
```

```
Sample paragraph defined.
```

✓ Count Words

Subtask:

Count the total number of words in the loaded text.

Reasoning: As per the subtask instructions, I will split the `sample_paragraph` by whitespace to get a list of words. Then, I will count the number of words in this list using the `len()` function and store the result in a variable named `word_count`. Finally, I will print the `word_count`.

Reasoning: Based on the reasoning provided, I will now implement the Python code to split the `sample_paragraph`, count the words, store the result in `word_count`, and print it.

```
words = sample_paragraph.split()  
word_count = len(words)  
print(f"Total word count: {word_count}")
```

```
Total word count: 56
```

Convert Text to Lowercase

Subtask:

Convert the entire `sample_paragraph` to lowercase.

✓ Convert Text to Lowercase

Subtask:

Convert the entire `sample_paragraph` to lowercase.

Reasoning: As instructed, I will convert the `sample_paragraph` to lowercase using the `.lower()` string method. The result will be stored in a new variable called `lowercase_paragraph` for further use and then printed to confirm the conversion.

Reasoning: Now that the reasoning for converting to lowercase is in place, I will write the Python code to perform the conversion using the `.lower()` string method, store the result in `lowercase_paragraph`, and then print the new lowercase string.

```
lowercase_paragraph = sample_paragraph.lower()
print(f"Lowercase paragraph:\n{lowercase_paragraph}")
```

```
Lowercase paragraph:
natural language processing (nlp) is a field of artificial intelligence that foc
```

✓ Summarize Preprocessing Steps

Subtask:

Summarize the preprocessing steps performed, including the total word count and the lowercase version of the text.

Reasoning: I will present a summary of the preprocessing steps, which includes the `word_count` and the `lowercase_paragraph` as requested in the subtask. These variables already contain the processed data.

Reasoning: Now that the reasoning for summarizing the preprocessing steps is in place, I will write the Python code to print the total word count and the lowercase version of the paragraph.

```
print("--- Preprocessing Summary ---")
print(f"Original paragraph word count: {word_count}")
print(f"Lowercase paragraph:\n{lowercase_paragraph}")
print("-----")
```

```
--- Preprocessing Summary ---
Original paragraph word count: 56
Lowercase paragraph:
natural language processing (nlp) is a field of artificial intelligence that foc
-----
```

Summary:

Data Analysis Key Findings

- NLTK and spaCy libraries were confirmed to be already present and were successfully imported into the environment.
- A sample paragraph was defined and utilized for subsequent preprocessing steps.
- The total word count of the sample paragraph was determined to be 56 words.
- The entire sample paragraph was successfully converted to its lowercase equivalent.
- A summary of the preprocessing steps was generated, including the calculated word count and the lowercase version of the text.

Insights or Next Steps

- The successfully executed preprocessing steps (word counting and lowercasing) lay the groundwork for more advanced Natural Language Processing (NLP) tasks.
- Further text preprocessing could involve tokenization, removal of stop words, stemming/lemmatization, or part-of-speech tagging using the imported NLTK and spaCy libraries.